BASE CHANGE FOR THE DERIVED UPPER SHRIEK OVER A CARTESIAN SQUARE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base change for the derived lower shriek over a cartesian square of quasi-compact quasi-separated schemes with f separated finite type: g^{-1} composed with Rf_! equals Rf'_! composed with (g')^{-1} (more-etale, Lemma lemma-base-change-shriek, canonical tag 0F7L). cited at proof line 4236. The single named input: the lower-shriek base change isomorphism from which the upper-shriek base change follows by uniqueness of adjoints.

Source reference. Original lemma 0F7L: sources/stacks-chow/tags.txt, source lines 19011–19011. Cited at proof line 4236: the lower-shriek base change isomorphism from which the statement follows by uniqueness of adjoints.

Lemma 0.2. *Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

of quasi-compact and quasi-separated schemes with f separated and of finite type. Then we have $Rf^! \circ Rg_* = Rg'_* \circ R(f')^!$.

Proof. By uniqueness of adjoint functors this follows from base change for derived lower shriek: we have $g^{-1} \circ Rf_! = Rf'_! \circ (g')^{-1}$ by Lemma 0F7L. □

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